

MARC LAVOIE

SOC ANALYST | SECURITY ENGINEERING | INCIDENT RESPONSE

Open to remote work and relocation

linkedin.com/in/marc-lavoie | mlavoie6927.github.io

15

MUNICIPAL NETWORKS
MONITORED

3.91

GPA | CUM LAUDE

15

RUNBOOKS FOR A 12-PERSON
LAB

1,538

ATTACK TECHNIQUES
CATALOGED

PROFESSIONAL PROFILE

Cybersecurity analyst with hands-on experience monitoring municipal networks, triaging alerts, tuning detections, and performing malware and digital-forensics investigations. Translates technical evidence into practical remediation, clear case notes, and repeatable playbooks. B.A.S. in IT Networking and Cybersecurity, 3.91 GPA, cum laude.

CORE SKILLS

Security Operations: Alert triage, incident response, threat hunting, detection tuning, vulnerability assessment, MITRE ATT&CK

SIEM & Telemetry: Elastic, Wazuh, Sentinel, QRadar, Falcon, Sysmon, Suricata, Zeek

DFIR & Malware: FTK Imager, Autopsy, Volatility 3, Wireshark, Security Onion, Any.Run, PeStudio

Systems & Automation: Windows, Linux, AD, VMware, pfSense, VyOS, Google Cloud, PowerShell, Python

PROFESSIONAL EXPERIENCE

SOC Analyst Apprentice - DFIR, Malware Analysis & Incident Response

MAR 2024 - PRESENT

LetsDefend

Remote | Full-time apprenticeship

- Investigate simulated incidents from alert validation through evidence capture, containment, eradication validation, and case documentation.
- Perform static and dynamic malware analysis with PeStudio and Any.Run; identify IOCs, persistence, process behavior, and network activity.
- Correlate QRadar offenses with FTK Imager and Autopsy evidence to separate actionable threats from benign signals and build defensible timelines.

Tier 1 SOC Analyst / IT Operations

MAR 2024 - DEC 2025

PISCES Program / PISCES International

Bellingham, WA | Full-time internship

- Monitored and defended 15 municipal networks with Elastic Stack and Suricata; triaged alerts and investigated suspicious network activity.
- Developed DNS-anomaly queries that identified rogue resolver behavior; tuned policies to reduce false positives and improve actionable visibility.
- Mapped CVEs to business impact and cost-effective mitigation roadmaps adopted by municipalities; reviewed peer detection queries and scripts.

GenCyber Mentor

SUMMERS 2024 & 2025

GenCyber Program

Bellingham, WA

- Taught cybersecurity, programming, and digital-forensics fundamentals; designed and quality-checked hands-on labs.
- Guided students through evidence-based troubleshooting and communicated complex concepts in clear, practical terms.

Network & Systems Administrator Intern

OCT 2024 - MAY 2025

Academic Cybersecurity Lab

Bellingham, WA | Hybrid/remote

- Built a VMware enterprise network with VyOS and pfSense segmentation, secure routing, perimeter defense, Windows hardening, and Active Directory controls.
- Authored 15 security playbooks and runbooks used by 12 teammates for consistent operations and incident handling.

ADDITIONAL EXPERIENCE

IT Support Specialist

Town of Dover

MAY 2022 - AUG 2022

Dover, MA | Contract

- Supported municipal staff with endpoint, account, and connectivity issues while coaching users on suspicious-email handling and secure workstation practices.
- Coordinated network restoration with third-party providers and documented hardware, software, and ArcGIS requirements for future upgrades.

IT Consultant

Independent Contract Work

FEB 1995 - APR 2022

Rhode Island | Part time

- Diagnosed Windows failures, malware infections, driver conflicts, and failing hardware for residential, education, and small-business clients.
- Hardened routers and endpoints with firmware updates, WPS removal, credential changes, port restrictions, layered protection, MFA, and safe-link guidance.

FEATURED SECURITY & ENGINEERING PROJECTS

SOC DECISION QUALITY

CrowdStrike True/False-Positive Investigations

Correlated Falcon alerts with Wireshark and Sysmon telemetry to prove one C2 case and disprove another while preserving the complete evidence trail.

FALCON | WIRESHARK | SYSMON

SECURITY ECOSYSTEMS

Project Atlas: Microsoft + CrowdStrike Stacks

Mapped end-to-end telemetry, identity, endpoint, data protection, detection, response, hunting, and governance workflows across two enterprise platforms.

SENTINEL | DEFENDER XDR | FALCON

NETWORK SECURITY

Service Provider NOC Security Engineering Lab

Designed a complete carrier topology with OSPF, BGP, MPLS L3VPN, VRFs, BFD, QoS, AAA, telemetry, validation commands, failure tests, and rollback guidance.

BGP | OSPF | MPLS | VRF | BFD

DEFENSIVE RESEARCH

2026 Master Cyberattack Taxonomy

Authored a 141-page visual reference spanning 50 attack domains and 1,538 named techniques with analyst focus, control priorities, and complete indexes.

50 DOMAINS | 1,538 ENTRIES

SIEM & DETECTION

Wazuh, ELK/Suricata & Cross-Platform Rules

Built Wazuh and ELK/Suricata labs, onboarded Windows telemetry, validated an approximately 500-command hardening baseline, and documented rules for Sentinel, QRadar, Chronicle, and Falcon.

WAZUH | ELASTIC | SURICATA | SIGMA

DFIR & AUTHORIZED TESTING

Forensics, Malware & Web Assessment Labs

Acquired and mounted forensic images, reconstructed malware timelines, and documented four isolated, instructor-provided web and Linux assessment workflows.

FTK | AUTOPSY | SECURITY UNION | NMAP

EDUCATION, CREDENTIALS & RECOGNITION

B.A.S., IT Networking & Cybersecurity

Whatcom Community College | 2025 | GPA 3.91 | Cum Laude | Dean's List

A.A.S., Information Technology & Network Engineering

New England Institute of Technology | 2021

SELECTED CREDENTIALS

- PISCES Analyst Professional | Sep 2025
- LetsDefend SOC Analyst + Malware Analyst | 2025
- Google Cybersecurity + Google Cloud | 2025
- 38 additional Cybrary and INE certificates
- PISCES Student of the Week recognition

COMPLETE PROJECT INVENTORY

Every completed project currently presented in the public portfolio is grouped below. Each section heading links directly to the supporting report, screenshot, video, or project evidence.

SOC OPERATIONS & DETECTION

[VIEW EVIDENCE](#)

- CrowdStrike Falcon True-Positive C2 Investigation
- CrowdStrike Falcon False-Positive C2 Investigation
- Sysmon Log Analysis
- Malware Investigation
- Remcos RAT Windows Investigation
- Cross-Platform Detection Rule Examples

NETWORK, CLOUD & PERIMETER

[VIEW EVIDENCE](#)

- Service Provider NOC Security Engineering Lab
- MPLS Layer-3 VPN Service Provider Lab
- Telecom + MPLS + 5G Cybersecurity Super Lab
- Firewall Policy + DDoS Defense
- Cyber Portfolio Lab
- Rackspace Security Labs

SIEM, ENDPOINT & HARDENING

[VIEW EVIDENCE](#)

- Wazuh Installation
- ELK Stack Lab with Suricata and Sysmon
- Windows Hardening Script Examples

PLATFORM ARCHITECTURE & RESEARCH

[VIEW EVIDENCE](#)

- Project Atlas: Complete Microsoft Security Stack
- Project Atlas: Full-Stack CrowdStrike Portfolio
- Bastion and Bean Security Architecture

DFIR & AUTHORIZED SECURITY TESTING

[VIEW EVIDENCE](#)

- FTK Image Acquisition and Mounting Analysis
- 2026 Master Cyberattack Taxonomy
- Authorized Web Security Simulation
- Linux Service Discovery & Enumeration
- MySQL & WordPress Security Assessment
- PHP & Apache Vulnerability Validation

SOFTWARE, AUTOMATION & WEB

[VIEW EVIDENCE](#)

- Credit Card Validator C++
- PowerShell Endpoint Inventory & Event Review
- Python Log Parser & Analyst Findings Report
- Responsive Cybersecurity Portfolio Website

PORTFOLIO WEBSITE ENGINEERING

Public evidence designed for recruiters, analysts, and engineering teams

Built and published a responsive GitHub Pages site with plain HTML, CSS, and JavaScript; organized evidence-heavy projects, mobile navigation, accessible interactions, direct PDF/video viewing, public-safe asset handling, and deployment verification.

[OPEN PORTFOLIO](#)